uc3m AIRBUS

Airframe Technology Master

Checkstress
Lecture 12. Metallic Sizing
(Tension Fittings / Attachments)

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10 June 2025

L12. References

1. Bruhn, E.F., “Analysis and Design of Flight Vehicle Structures”, Jacobs Publishing Inc., 1973
2. M. C. Y. Niu, “Airframe Stress Analysis and Sizing”
3. HSB21020-01, “Procedure for the determination of the static joint strength values of mechanically fastened metal

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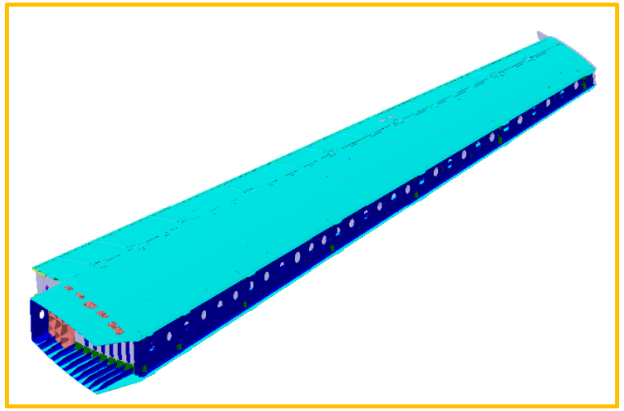
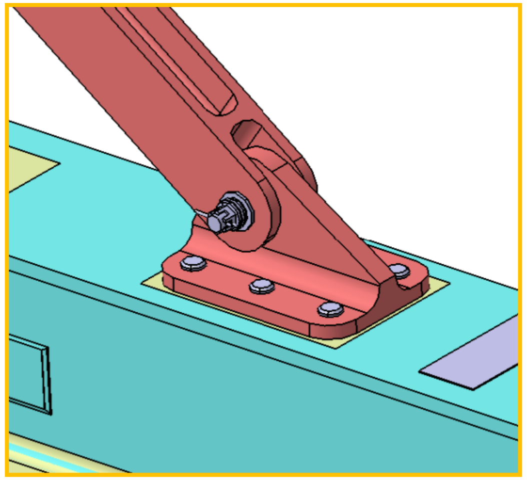
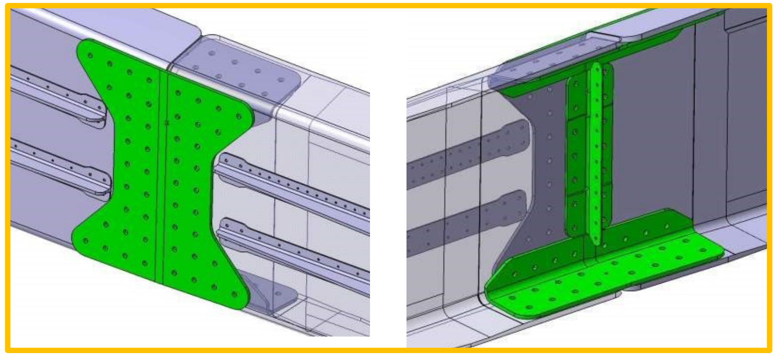
Joint Definition

A joint is an assembly of two or more plates fixed together by fasteners (bolts or rivets).

The purpose of a joint is to provide a load path between different parts.







Joint Definition

A joint is defined by the following parameters:

- Part materials: nature, heat treatment, thickness billet
- Geometry: type of joint, edge distances, pitches, etc…
- Fastener systems: type of bolt/rivet and nut/collar, material, geometry, preload, washer presence, etc…

Part Materials:

Requirements from certification agencies (EASA, FAA):

- Material strength properties must be based on enough tests of material meeting approved specifications to establish design values on a statistical basis.
- Design values must be chosen to minimize the probability of structural failures due to material variability.
- Two probability classes, depending on the structural safety concept being used:
  - **A-value:** Design value determined with a probability of 99% and confidence level of 95%. Used for single load path parts.
  - **B-value:** Design value determined with a probability of 90% and confidence level of 95%. Used for multiple load paths parts.

The material design values required for the analysis are dependent on the heat treatment and the billet thickness of the material.

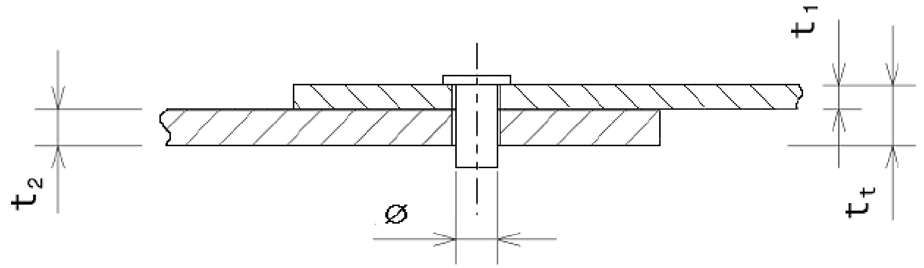
Joint Definition

Geometry:

— TYPE OF JOINTS:

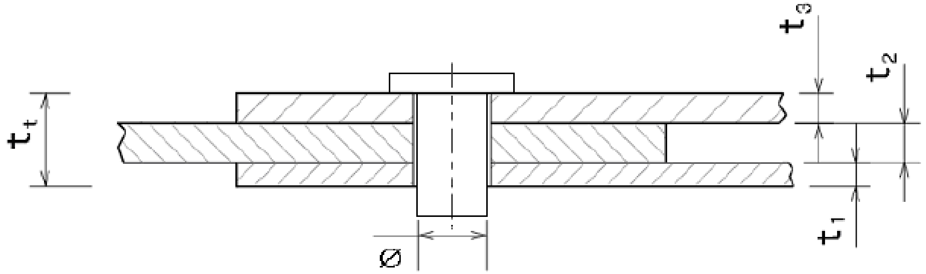
SLS - Single Lap Shear Joint:

Joints with two overlapping plates and with only one shear plane in the fastener



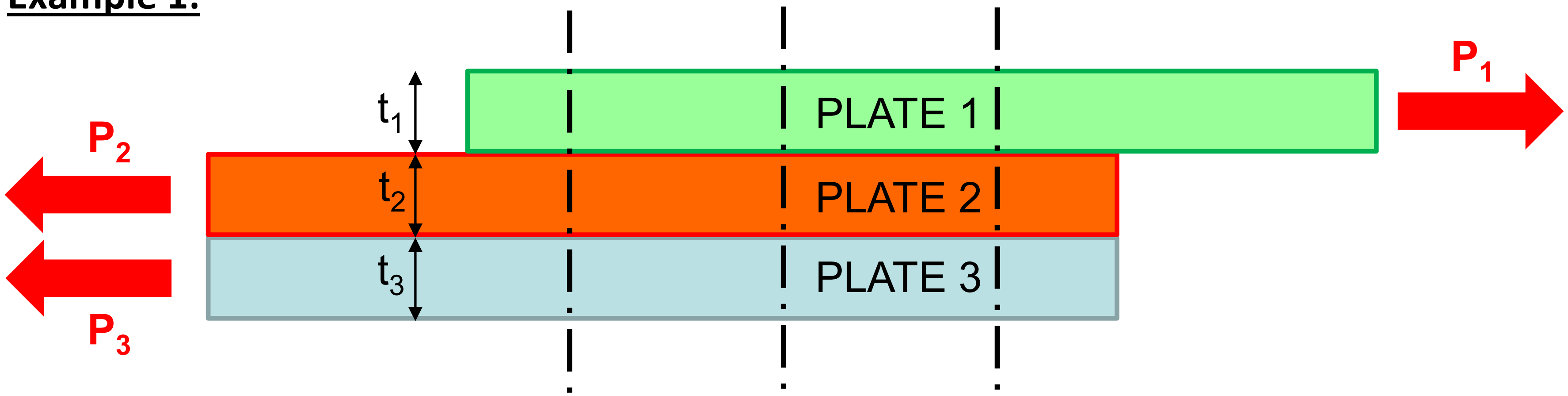
DLS – Double Lap Shear Joint:

Joints with more than two plates and several shear planes in the fastener

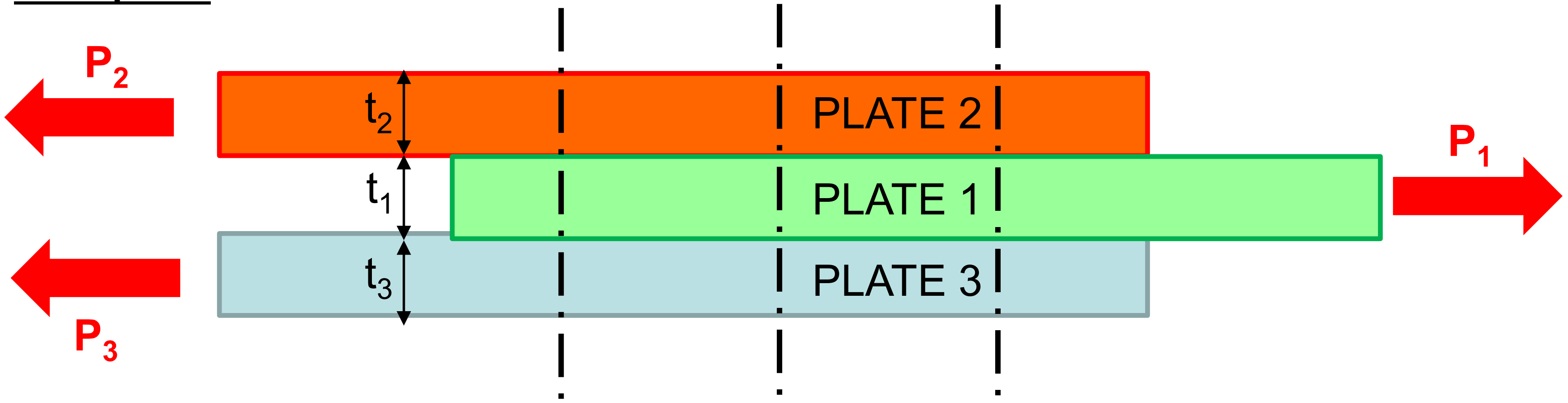


Joint Definition

Example 1:



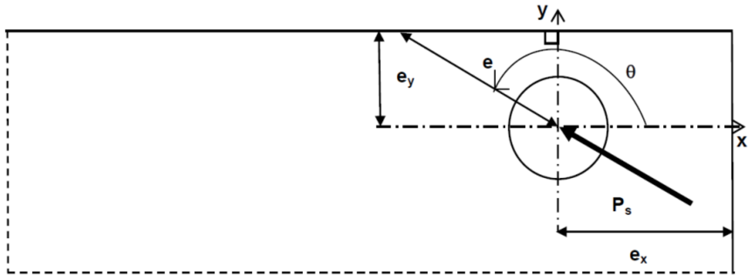
Example 2:



Geometry (cont):

— EDGE DISTANCE:

The edge distance is the distance between the centre of the fastener hole and the edge of the plate (in the load direction). For metallic joints, typical edge distance is e≥2D.

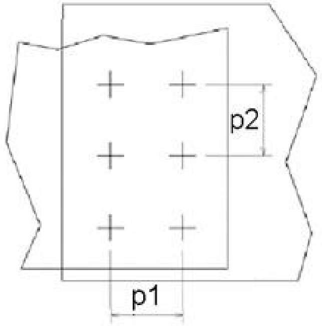


Geometry (cont):

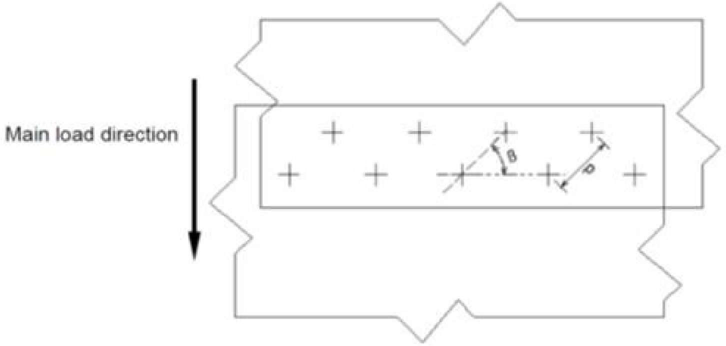
— <u>PITCH:</u>

Distance between the centre of two adjacent fasteners. For metallic joints, typical pitch are in the range 4.5D ≤ p ≤ 6D. A reduced pitch of 4D can be used for larger diameter.

Rectangular patern



Staggered hole patern



Geometry (cont):

— t/D:

To reach the maximum capability of the joint, plate thickness over diameter ratio is normally between $0.5<t/D<2.0$

- If $t/D<0.5$, failure is driven by bearing of the plate
- If $t/D>2.0$, failure is driven by fastener shear strength (maximum strength capability of the attachment)

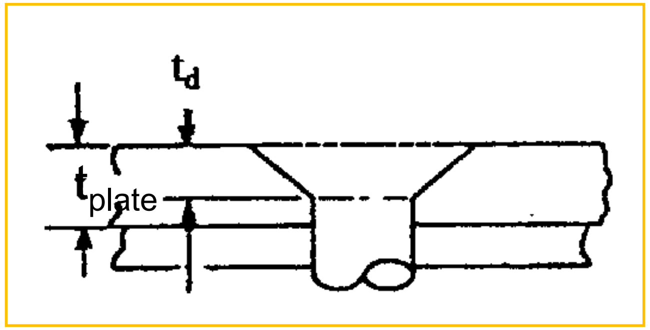
Fastener System:

— Countersunk fasteners:

To avoid the knife edge effect: $\boxed{t_{plate}/t_d > 3/2 \rightarrow t_d < (2/3)\cdot t_{plate}}$

Where **t_{plate} (mm)**, is the plate thickness

t_d (mm) is the countersunk head height

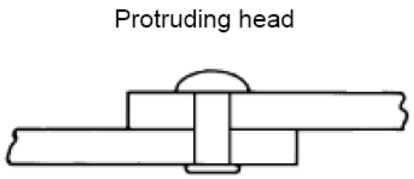


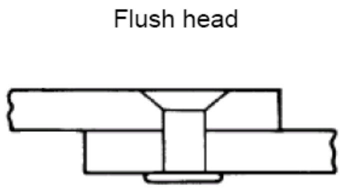
Joint Definition

Fastener System (cont):

— SOLID RIVETS:

- Defined as one-piece fasteners installed by mechanically deforming one end.
- Main characteristics: Permanent fasteners, low cost fastener, low cost assembly (suitable for automatic installation).
- Mechanical properties: Limited static shear strength (materials and diameters used), bad static strength in tension (out of plane), good fatigue strength in shear (solid rivets fill up the hole).
- Typical use: Appropriate when non-removable cheap fasteners needed, on areas with moderate static shear loads only.





Joint Definition

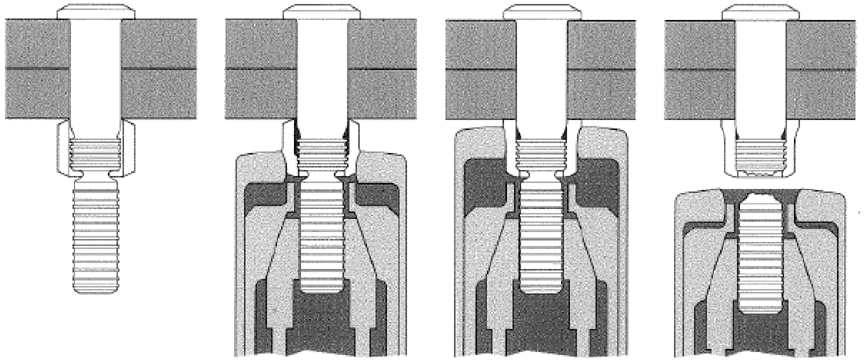
Fastener System (cont):

— LOCKBOLT FASTENERS:

- Swaged collar close-tolerance fasteners designed mainly for shear joints. A Lockbolt consists of a solid pin and a malleable collar, which is swaged onto the grooves of the pin to clamp the joint.
- Main characteristics: Permanent fasteners, close-tolerance fastener, moderate manufacturing cost of the fastener, low cost assembly (suitable for automatic installation), sufficient space is required for installation.
- Mechanical properties: Relative high static shear strength, good fatigue strength in shear, moderate static and fatigue strength in tension when tension-type Lockbolt used, low if shear or medium type used.
- Typical use: Appropriate when non-removable shear fasteners are needed on common structural areas, for high volume production, with sufficient space for installation and with limited static tensile loads.

Fastener System (cont):

— LOCKBOLT FASTENERS:



1 The pin is inserted from one side while the collar is placed.	2 The pin is pulled while the collar material is swaged into the pin grooves, providing a permanent lock between the pin and the collar.	3 Once the collar swaging is complete, the pintail separates, at a predetermined force.

Joint Definition

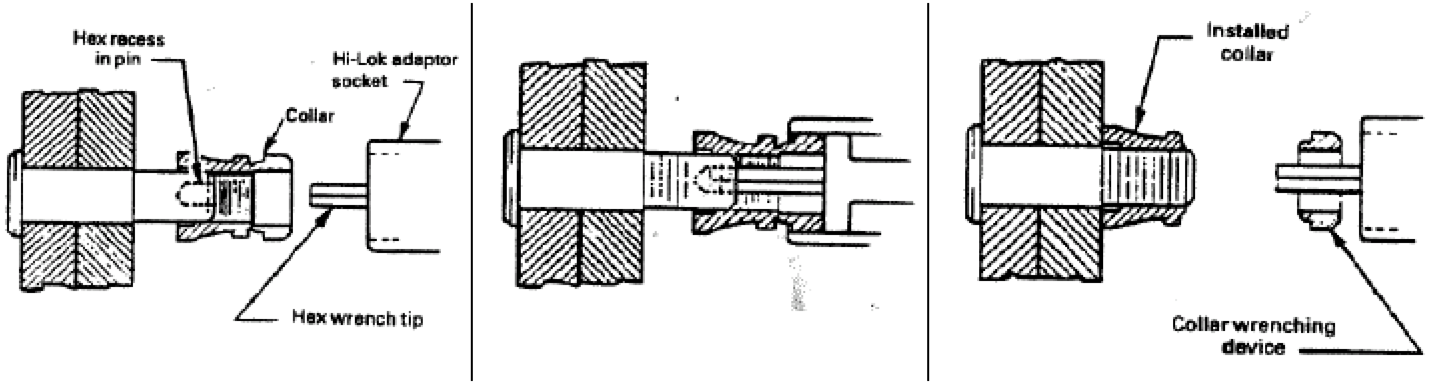
Fastener System (cont):

— HI-LOK, HI-LITE FASTENERS:

- Close-tolerance threaded fasteners optimized for shear joints (also have tension capability). Consists on a threaded precision pin and a threaded collar (nut). The breakable part of the collar automatically shear off at a predefined torque level that leaves the installation with the correct amount of torque and preload.
- Hi-Lite: Optimization of Hi-Lok system to reduce weight. Similar mechanical properties.
- Main characteristics: In general, similar to Lockbolts systems, moderate assembly cost (not adapted to high volume production as solid rivets or Lockbolts), available in large diameters for very high shear loads.
- Mechanical properties: High static shear strength, good fatigue shear strength due to good pre-load and close-tolerance, static tension strength generally driven by the nut.
- Typical use: Good alternative to Lockbolts when not enough space for Lockbolt installation or very high shear strength required (large diameters).

Fastener System (cont):

— <u>HI-LOK, HI-LITE FASTENERS:</u>



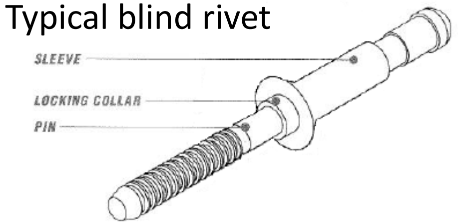
Joint Definition

Fastener System (cont):

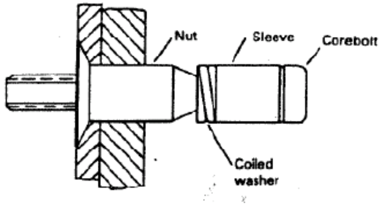
— BLIND FASTENERS (BLIND BOLTS AND BLIND RIVETS):

- Fastener that can be completely installed from only one side of the structure.
- Installation of blind rivets performed by pulling the pin through the sleeve. Installation of blind bolts performed by rotation of the pin.
- Main characteristics: Low cost assembly (suitable for automatic installation).
- Mechanical properties: Relative good static shear strength, low fatigue strength in shear and tension.
- Typical use: Due to low fatigue strength, its use is strictly limited to joints where the access to both sides is not possible. Often used as temporary solution for repairs.

Typical blind rivet



Typical blind bolt



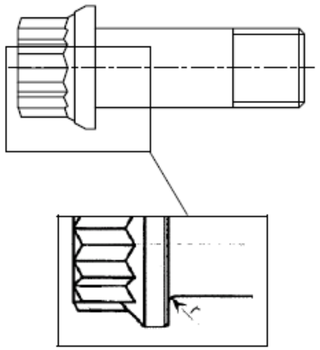
Fastener System (cont):

— TENSION BOLT:

- Threaded protruding head fasteners designed and optimized for fatigue tension predominant loads: pure tension or tension combined with shear load cases. Tension bolts have enlarged root radius under the head to improve tension/bending fatigue life.

- Main characteristics: high manufacturing cost, relatively high assembly cost, not adapted to high volume production.

- Mechanical properties: High static shear strength, very high static tension strength caused by the high core section in tension, high tension fatigue strength if used with high pre-load, acceptable fatigue strength in shear if used with high pre-load.

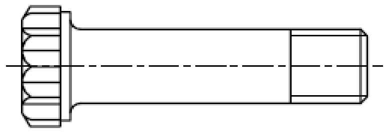


- Typical use: Recommended only on structure where static tension is the primary loading or when high fatigue tension strength is required.

Fastener System (cont):

REMOVABLE SHEAR BOLTS:

- Threaded fasteners designed for shear application. Consists on a pin with a short head and a thread with its associated nut.
- Main characteristics: Same as Tension bolts except lower weight and lower pre-load capability.
- Mechanical properties: High static shear strength, moderate static tensile strength (limited by the associated shear nut), moderate tensile fatigue strength caused by the moderate value of the pre-load, acceptable fatigue strength in shear if installed with slight clearance.
- Typical use: To use on shear joints with moderate tension loads where removable fasteners are explicitly required.



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CS 25.625 Fitting factor

For each fitting (a part or terminal used to join one structural member to another), the following apply:

(a) For each fitting whose strength is not proven by limit and ultimate load tests in which actual stress conditions are simulated in the fitting and surrounding structures, a fitting factor of at least 1,15 must be applied to each part of –

(1) The fitting;

(2) The means of attachment; and

(3) The bearing on the joined members.

(b) No fitting factor need be used –

(1) For joints made under approved practices and based on comprehensive test data (such as continuous joints in metal plating, welded joints, and scarf joints in wood); or

(2) With respect to any bearing surface for which a larger special factor is used.

(c) For each integral fitting, the part must be treated as a fitting up to the joint at which the section properties become typical of the member.

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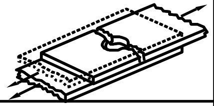
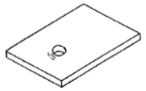
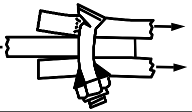
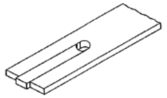
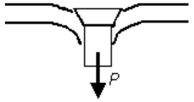
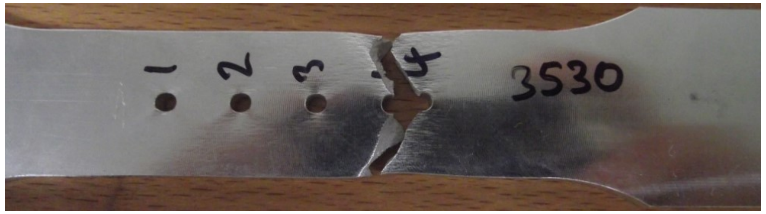
Failure Modes

Bolt loading	Failure mode	Critical element
Shear	Net section 	Plate
	Bolt 	Fastener System
	Bearing 	Plate
	Bolt / bearing 	Plate+Fastener System
	Shear out 	Plate
Tension	Bolt/Nut 	Fastener System
	Pull-through 	Plate
Combination Shear + Tension	Bolt 	Fastener System

Failure Modes

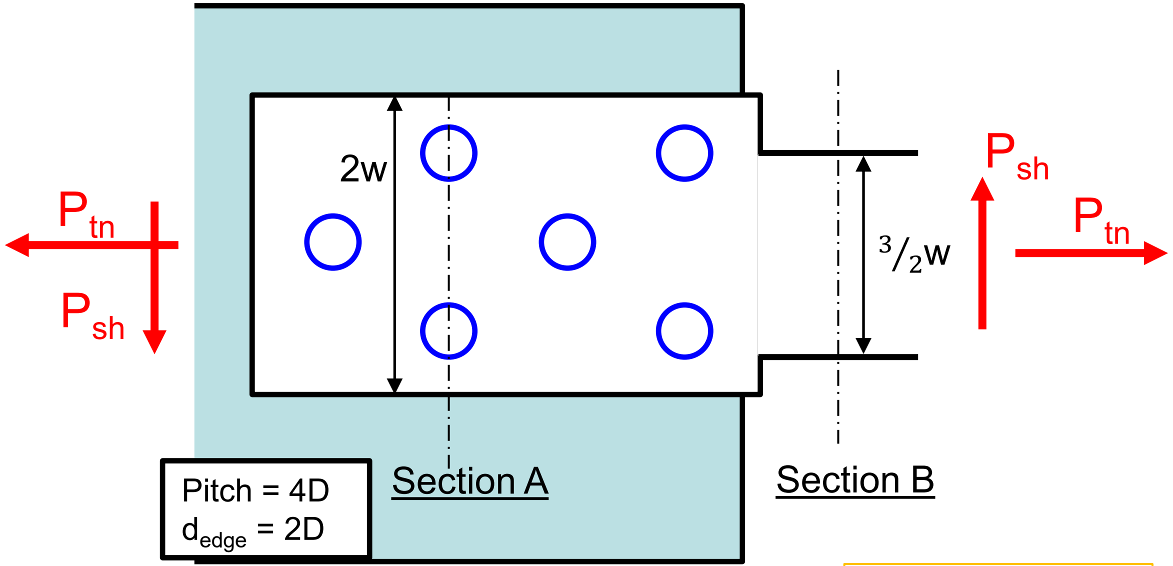
Shear Loads:

— <u>NET SECTION FAILURE:</u>



- Ultimate allowable of the plate considering the presence of the holes. Calculation to be done on the minimum section area (A_{net} = net area). Net area calculation shall be corrected in case of countersunk holes.

Which section is the most critical? Note: assume constant thickness and protuberant bolts



$$l_{net\_A} = 2w - 2D = 2w - {}^{w}/_{2} = {}^{3}/_{2}\, w$$

$$l_{net\_B} = {}^{3}/_{2}\, w$$

Shear Loads (cont):

— NET SECTION FAILURE (cont):

- Determination of the applied stress at critical section:

For tensile loads: $\sigma_{tn} = \frac{P_{tn}}{A_{net}}$

For shear loads: $\tau_{applied} = \frac{P_{sh}}{A_{net}}$

- Reserve Factor is:

$$\boxed{RF = \frac{1}{FF \cdot \sqrt{R_{tn}^2 + R_{sh}^2}}}$$

Where,

$$\mathrm{R_{tn}} = \frac{\sigma_{\mathrm{tn}}}{\mathrm{F_{tu}}};\ \mathrm{R_{sh}} = \frac{\tau_{\mathrm{applied}}}{\mathrm{F_{su}}}$$

Being **Ftu** = allowable tension stress of the plate material
Fsu = allowable shear stress of the plate material
FF = Fitting factor

Failure Modes

Shear Loads (cont):

— <u>BOLT FAILURE:</u>

- Maximum shear load that the sank of the bolt can withstand before failure.

- Usually, this value is provided directly by the supplier in the fastener standard. The value is provided as a force and corresponds to DLS configuration.
  →DLS configuration: $\mathrm{F_{shear\_bolt\_DLS}}$ from standard.
  →SLS configuration: $\mathrm{F_{shear\_bolt\_SLS}} = \frac{1}{2} \cdot \mathrm{F_{shear\_bolt\_DLS}}$

- When such information is not available in the standard (common case for solid rivets), the value can be estimated based on the shear strength of the material of the bolt:
  →DLS configuration: $F_{shear\_bolt\_DLS} = 2 \cdot Fsu \cdot \pi \cdot \frac{D^2}{4}$
  →SLS configuration: $F_{shear\_bolt\_SLS} = Fsu \cdot \pi \cdot \frac{D^2}{4}$

Shear Loads (cont):

— <u>BOLT FAILURE (cont):</u>

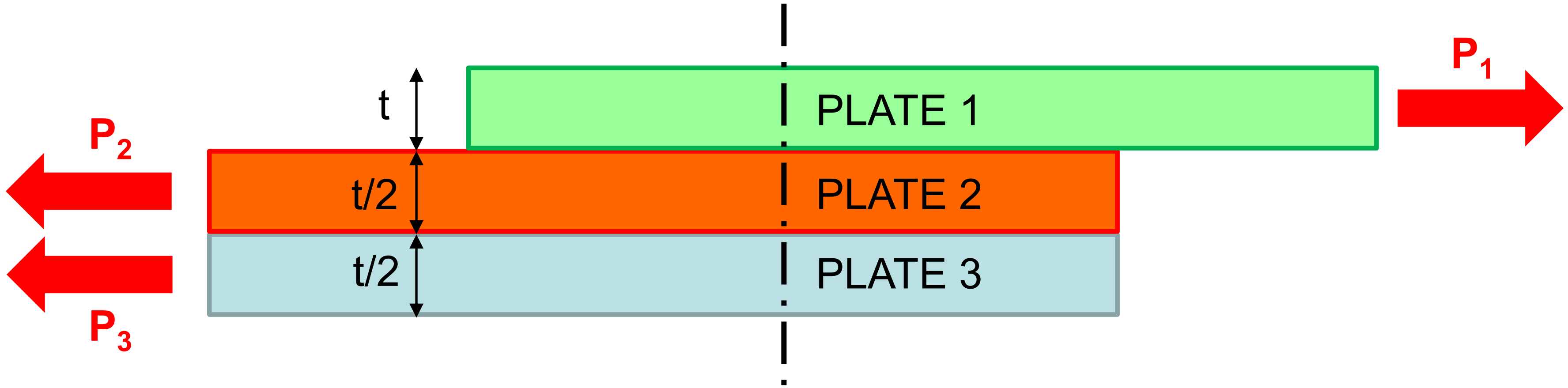
- Determination of RF by comparison with the applied shear force of the fastener.
- In case of a joint configuration with several plates, analysis shall be done at each joint interface.

$$\boxed{RF = \frac{F_{shear\_bolt\_\text{allowable}}}{\text{FF} \cdot F_{shear\_applied}}}$$

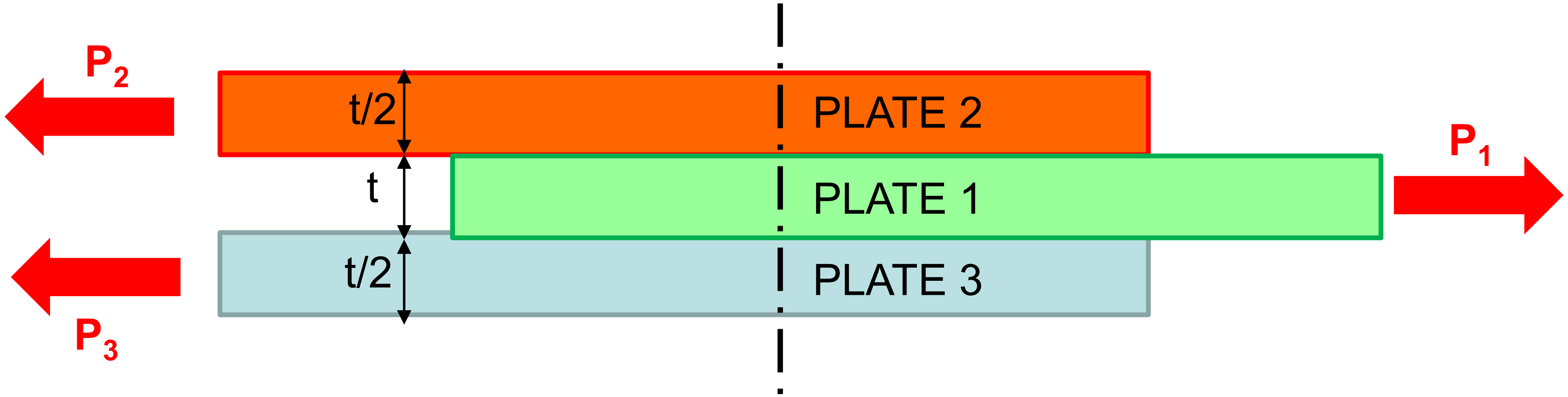
Joint Definition

Calculate the RF_{sh} if $F_{shear\_bolt\_DLS} = P_1$

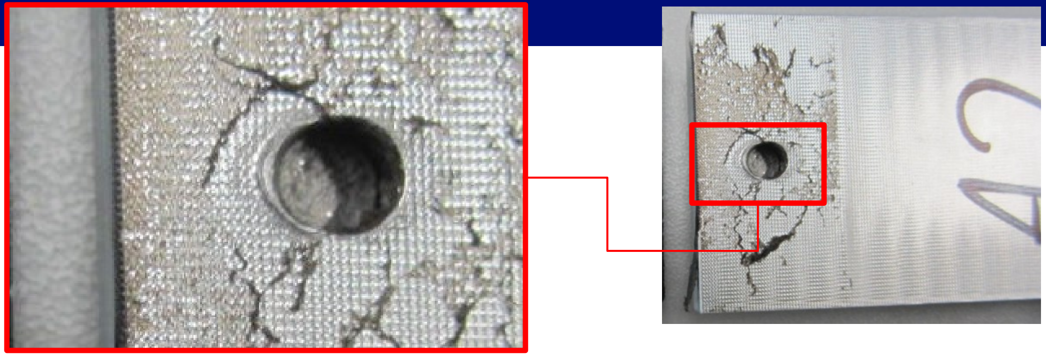
Example 1:



Example 2:



Failure Modes

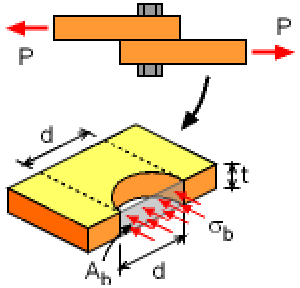


Shear Loads (cont):

— BEARING FAILURE:

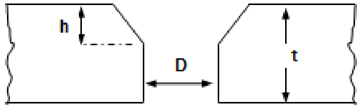
- Deformation on the hole caused by the transferred load through the fastener. The bearing strength of a material is obtained by specific tests, and by definition corresponds to the force acting on the hole divided by the projected area (A_{br}):

$$\sigma_{bearing\_applied} = \frac{P}{A_{br}}$$



→Protruding bolts: $A_{br} = D \cdot t$

→Countersunk bolts: $A_{br} = D \cdot (t - h) + \alpha \cdot D \cdot h$



→Solid rivets: $\alpha = \frac{2}{3} \cdot \frac{F_{su}}{F_{bru}} + 0.2$

→Hi-Lok, Hi-Lite, Lockbolt and Blind Rivets: $\alpha = \frac{2}{3} \cdot \frac{F_{su}}{F_{bru}}$

Shear Loads (cont):

— BEARING FAILURE (cont):

- Determination of RF by comparison with the applied bearing force transmitted by the fastener.

$$\boxed{RF = \frac{F_{br}(^{e}/_{D})}{FF \cdot \sigma_{bearing\_applied}}}$$

Where,

Fbr (e/D): Bearing allowable stress, $F_{br} = \min(F_{bru}(e/D), 1.5 \cdot F_{bry}(e/D))$

→Ultimate bearing Fbru is the maximum stress withstood by a bearing specimen

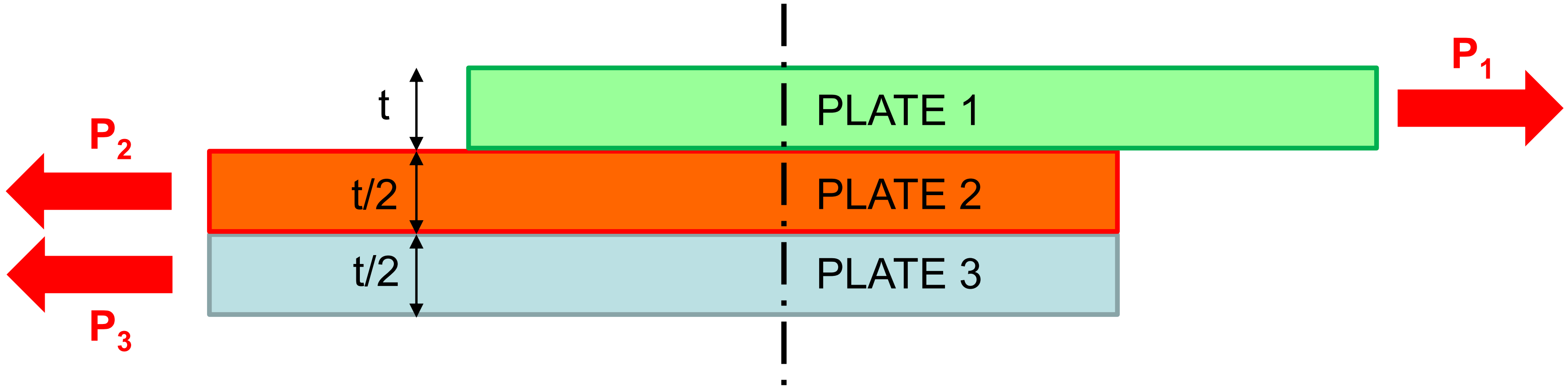
→Yield bearing Fbry by definition is the stress that causes a permanent deformation on the hole of 0.02·D

→Values determined by tests for two edge distances: e/D=1.5 and e/D=2.0. For a given configuration with e/D≥2.0, the value given at e/D=2.0 shall be used. Linear interpolation between both values for any 1.5<e/D<2.0. Specific tests shall be done to determine bearing strength below e/D<1.5.

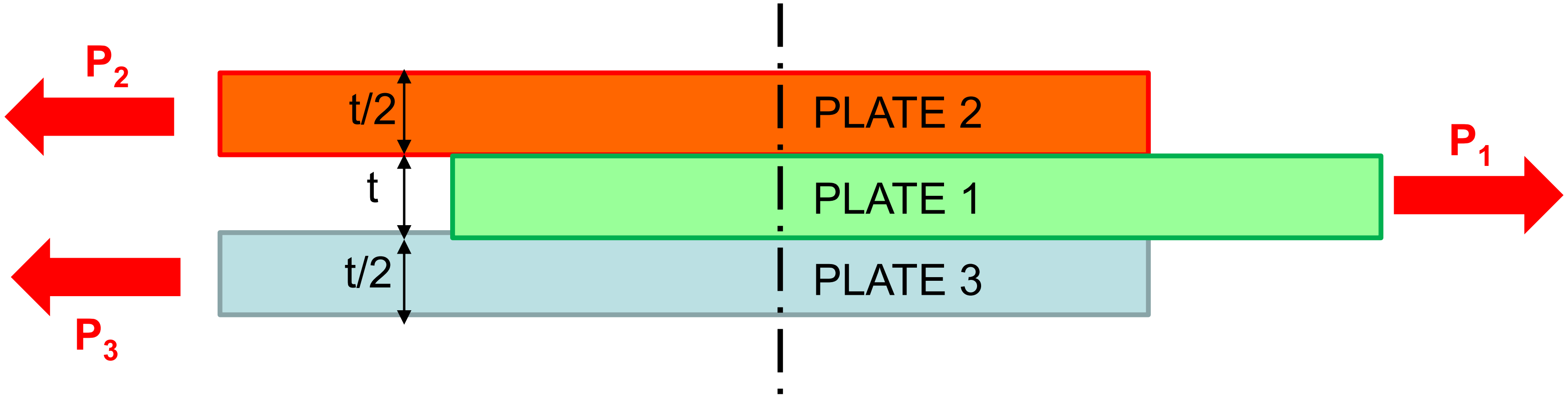
Joint Definition

Which plate is the most critical for bearing failure?

Example 1:



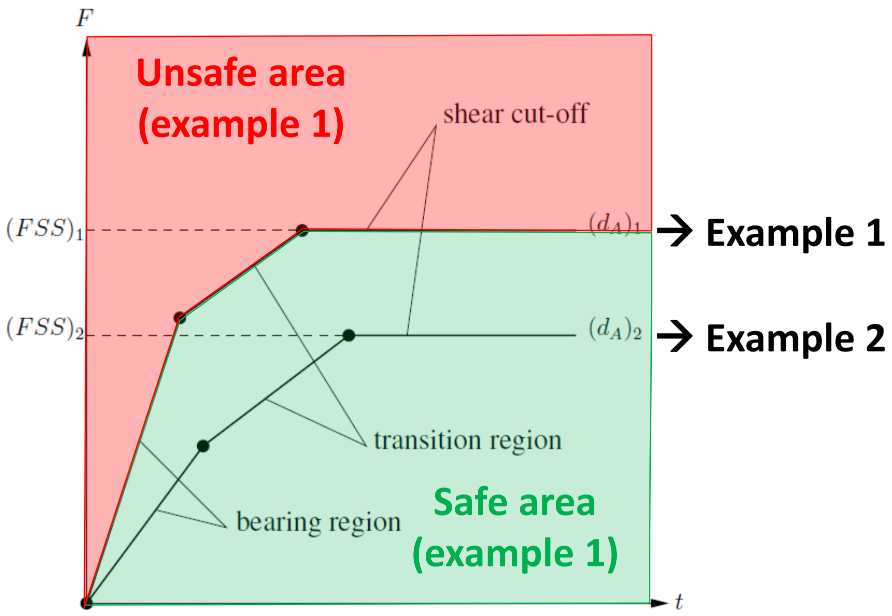
Example 2:



Shear Loads (cont):

BOLT/BEARING FAILURE:

- Combination of bearing failure in the plate and bolt failure. In general, the ultimate strength of a single shear joint can be represented by a three regions failure diagram.



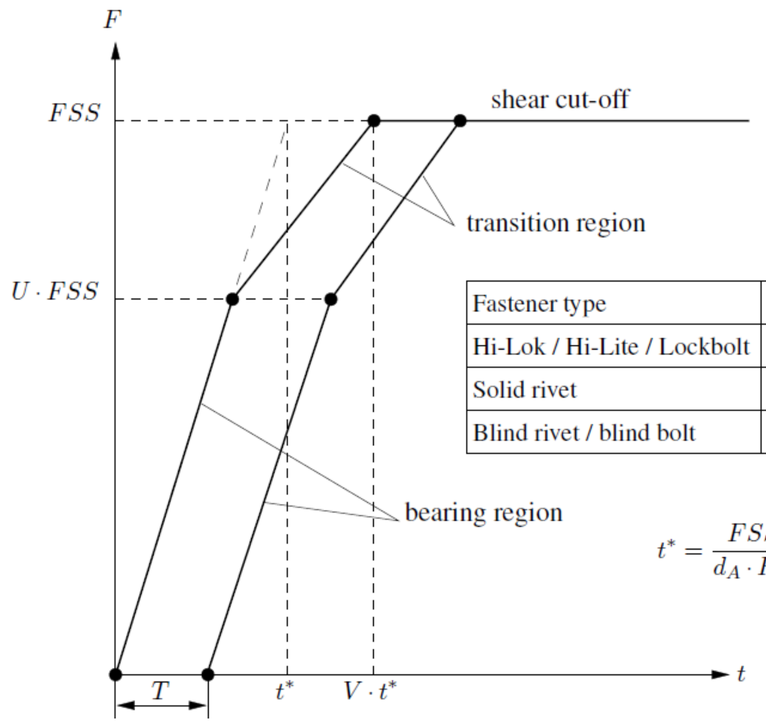
FSS: Fastener shear strength
d: Fastener diameter
t: Plate thickness

Failure Modes

Shear Loads (cont):

— BOLT/BEARING FAILURE (cont):

- **Protruding** head fasteners:



Fastener type	U [–]	V [–]	T [mm]
Hi-Lok / Hi-Lite / Lockbolt	0.85	1.3	0
Solid rivet	0.75	1.5	0
Blind rivet / blind bolt	0.60	1.7	$0.05 \cdot d_A$

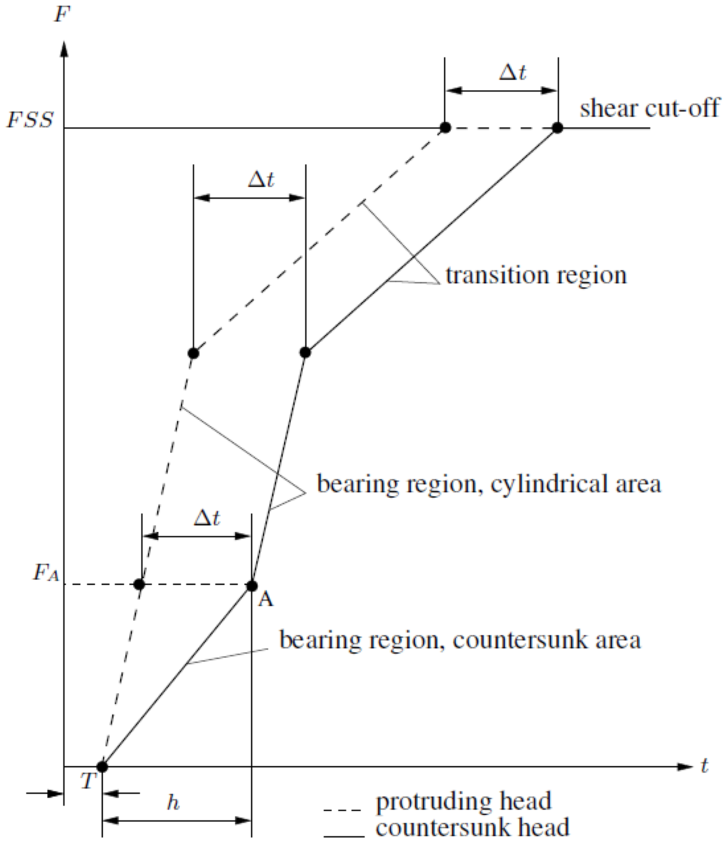
$$t^* = \frac{FSS}{d_A \cdot R_{br}}$$

FSS: Fastener shear strength
d: Fastener diameter
t: Plate thickness

Shear Loads (cont):

—— BOLT/BEARING FAILURE (cont):

- **Countersunk** head fasteners:

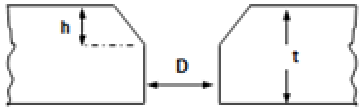


FSS: Fastener shear strength

Δt: Δt=(1-α)·h

- Solid rivets: $\alpha = \frac{2}{3} \cdot \frac{F_{su}}{F_{bru}} + 0.2$
- Hi-Lok, Hi-Lite, Lockbolt and Blind Rivets:

$$\alpha = \frac{2}{3} \cdot \frac{F_{su}}{F_{bru}}$$



Shear Loads (cont):

— BOLT/BEARING FAILURE (cont):

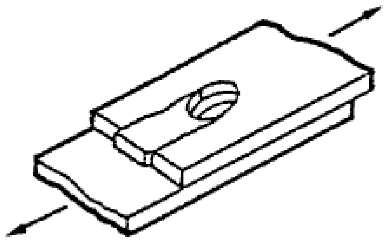
- Determination of RF by comparison between the applied force and the joint allowable in each side of the joint (i.e. head side and nut side):

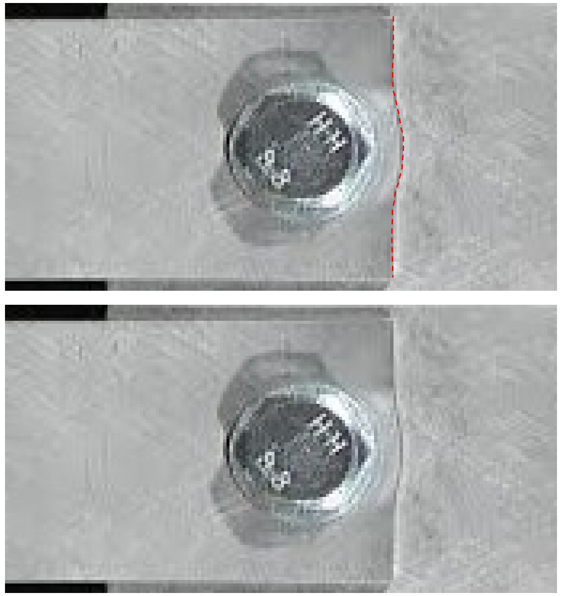
$$RF_{bolt/bearing} = \frac{F_{bolt/bearing\_allowable}}{FF \cdot F_{applied}}$$

Shear Loads (cont):

— SHEAR-OUT FAILURE:

- Failure of the metal plate due to the shear stresses around the hole. Can be considered as a particular case of bearing failure when the edge distance is reduced.
- Appears for configurations with low edge distance, values of $e/D<1.5$ are prone to this failure type. Can be avoided assuring $e/D \geq 2.0$ (typical design criteria of joints).





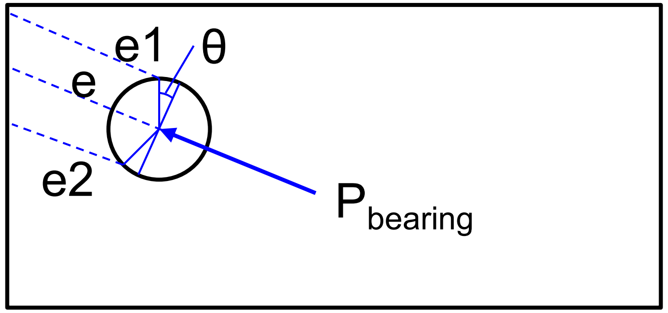
Shear Loads (cont):

— SHEAR-OUT FAILURE:

- Determination of RF by comparison between the applied shear force and the ultimate shear allowable of the material.

  →No calculation needed for e/D≥2.0, because other failure modes (i.e. bearing or fastener shear failure) becomes critical.
  →Determination of e1 & e2 at an angle θ from bearing load direction.

$$RF_{Shear\_out} = \frac{F_{su}}{\mathrm{FF} \cdot \tau_{applied}} = \frac{F_{su}}{\mathrm{FF} \cdot P_{bearing} \cdot A_{shear}} = \frac{F_{su}}{\mathrm{FF} \cdot P_{bearing} \cdot t \cdot (e_1 + e_2)}$$



Failure Modes

Tension Loads:

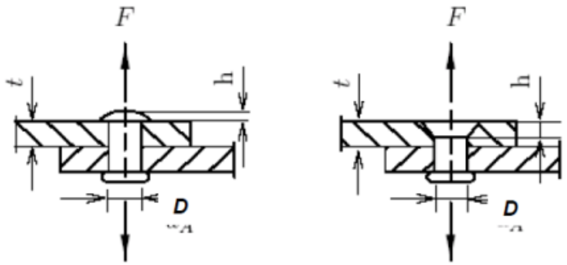
— BOLT FAILURE:

HEAD FAILURE

NUT FAILURE

- Maximum tension load that the bolt can withstand before failure.
- Usually, this value is provided directly by the supplier in the fastener and nut standards. The value is provided as a force.
- When such information is not available in the standard (common case for solid rivets), the value can be estimated based on the shear and tensile strength of the material of the bolt:

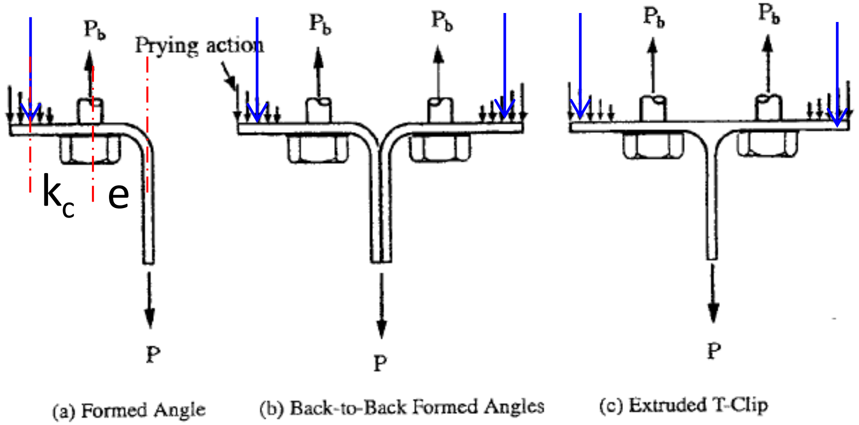
$$F_{tensile\_bolt} = min\left(Ftu \cdot \pi \cdot \frac{D^2}{4} ; \pi \cdot D \cdot h \cdot Fsu\right)$$



Tension Loads (cont):

— BOLT FAILURE (cont):

- **Prying factor shall be taken into account** when eccentric loads are applied on the fastener.



$P_{bolt} = \mathrm{P} \cdot C_{prying}$; where $C_{prying} = 1 + \frac{\mathrm{e}}{kc}$

Being e= Distance between bolt shank and mid of the web

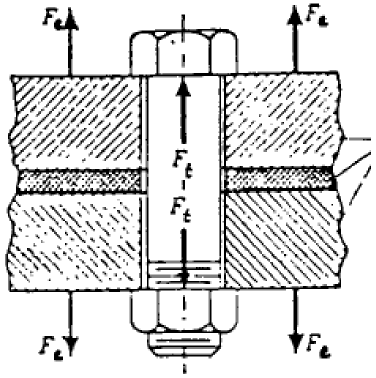
kc = Distance between the bolt shank and the equivalent prying load application.kc usually determined by tests, conservatively, a triangular distribution can be assumed

Tension Loads (cont):

— <u>BOLT FAILURE (cont):</u>

- Effect of preload in bolted joints.

  →When no external load is applied, only the contact pressure between the plates react the bolt preload.



  → When applying an external tensile load on the plates, only part is transferred to the fastener, the remaining force reacts the compressive state created by the preload. Once the plates are fully separated, the fastener carries the whole external load.

Tension Loads (cont):

— BOLT FAILURE (cont):

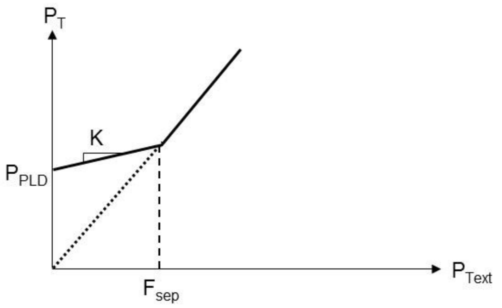
- Effect of preload in bolted joints.

→At ultimate load, the plates are considered separated, so the fastener system shall withstand the fully load. For low load levels (typical for fatigue calculations), the load in the fastener can be calculated as follows:

$$P_{applied\_bolt} = P_{PLD} + \frac{n \cdot kb}{kb + kf} \cdot P_{ext}$$

Being:

n: Loading plane factor. Usually n=0.5
kb: Bolt stiffness
kf: Plate stack compression stiffness



Tension Loads (cont):

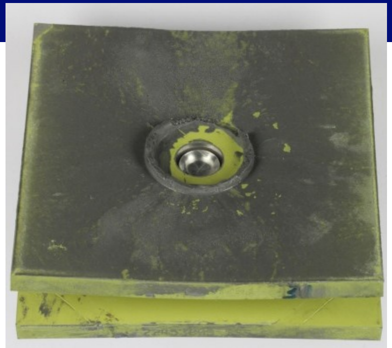
BOLT FAILURE (cont):

- Determination of RF by comparison of the applied tensile load and the tensile allowable of the fastener system. The minimum between the head (i.e. fastener tensile allowable) and the nut (i.e. nut tensile allowable) side of the joint shall be considered.

$$\boxed{RF_{Tensile\_head} = \frac{F_{tensile\_bolt}}{\mathrm{FF}\cdot P_{applied\_bolt}}}$$

$$\boxed{RF_{Tensile\_nut} = \frac{F_{tensile\_nut}}{\mathrm{FF}\cdot P_{applied\_bolt}}}$$

$$RF_{Tensile} = min\left(RF_{Tensile\_head}; RF_{Tensile\_nut}\right)$$



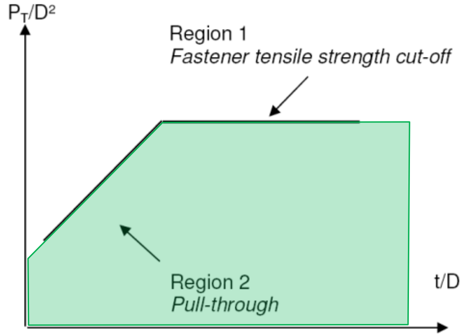
Tension Loads (cont):

— <u>PULL-THROUGH FAILURE:</u>

- Similarly as happens with shear loads, the behavior of a joint submitted to tensile loads is determined by two regions:

→Fastener tensile cut-off region, for high t/D where the joint is limited by the maximum tensile allowable of the fastener system.

→Pull-through region, for small t/D where the plate is critical due to earlier bending deformations around the hole area (with possibility of head or nut going through the plate thickness). In this case, the joint does not develop the tensile strength of the fastener system and the joint is limited by a pull-through failure of the metallic plates.



P_T: Applied tensile load

D: nominal diameter of the fastener

t: thickness of the sheets

Tension Loads (cont):

— PULL-THROUGH FAILURE (cont):

→Pull-through stress: $$\mathrm{f}_{su} = \frac{P}{0{,}6 \cdot \pi \cdot D_w t_e}$$

Where,

P: Applied tension load

D_w: Outside diameter of the washer or nut or head bolt

t_e: End thickness

Assuming 60% of effectivity

- Determination of RF by comparison of the applied tensile load and the pull-through allowable of the joint.

$$\boxed{RF_{Pull-through} = \frac{F_{\mathrm{su}}}{\mathrm{FF} \cdot f_{su}}}$$

Where,

F_{su}: Ultimate shear allowable

Combined Shear and Tension Loads:

- Only when the joint configuration is critical due the fastener system (i.e. limited by fastener system shear and tensile allowable), an interaction law for combined states are defined.
  - The interaction law are determined by tests and depends on the type of fastener system.
  - Typical interaction law as follows:

$$\boxed{RF_{bolt} = \frac{1}{FF\sqrt{R_t^2 + R_s^2}}}$$

Where:

$$R_t = \frac{P_{appliead\_tension}}{F_{tensile\_fastener-system}}$$

$$R_s = \frac{P_{appliead\_shear}}{F_{shear\_fastener-system}}$$

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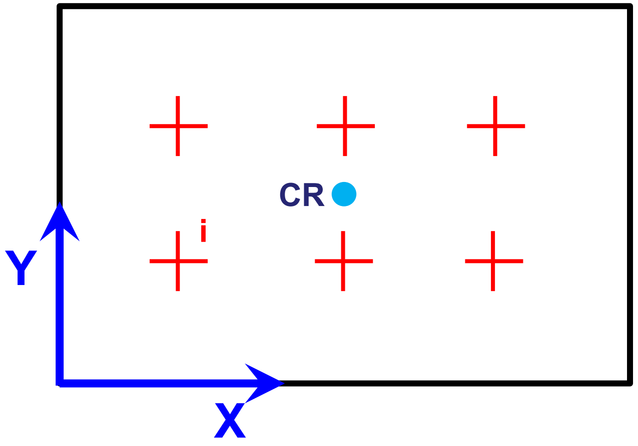
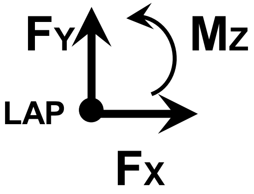
Bolt group approach:

Analytical determination of the loads acting on a group of bolts, assuming the whole group pivot around its centroid to react the applied loads without deformation of the joint.





INPUTS:
LAP = Load Aplication Point: (X_A, Y_A) (mm)
Applied loads: Fx (N), Fy (N), Mz (Nmm)
Distribution of bolts: (X_i, Y_i) (mm)

1/5 — Bolt group centroid calculation:

$$X_{CR} = \frac{\sum x_i \cdot A_i}{\sum A_i} \qquad Y_{CR} = \frac{\sum y_i \cdot A_i}{\sum A_i}$$

Bolt group approach (cont)

2/5 — Bolt group moment of inertia calculation:

$$I_x = \sum(Y_i - Y_{CR})^2 \cdot \mathrm{A}_i;\ I_y = \sum(X_i - X_{CR})^2 \cdot \mathrm{A}_i;\ I_p = I_x + I_y$$

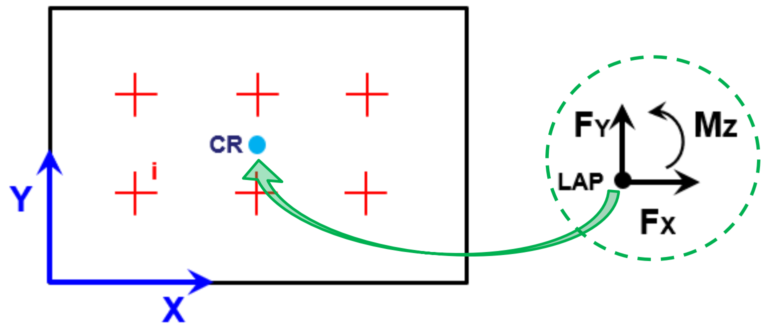
Where,

Xi; Yi (mm): Bolt coordinates

X_{CR}; Y_{CR} (mm): Center of riveting coordinates (From step 1/5)

3/5 — Forces and moments in the centroid:

$$\boxed{F_{x\_CR} = F_x} \quad \boxed{F_{y\_CR} = F_y} \quad \boxed{M_{z\_CR} = M_z - F_x \cdot (Y_A - Y_{CR}) + F_y \cdot (X_A - X_{CR})}$$



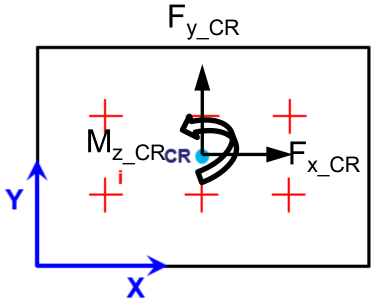
Bolt group approach (cont): INPLANE CALCULATION → FX, FY & MZ

4/5 —— Shear forces in a given bolt i:

Due to shear forces: $P_{x\_i}^{F_{x\_CR}} = \frac{F_{x\_CR} \cdot A_i}{\sum A_i}$ $P_{y\_i}^{F_{y\_CR}} = \frac{F_{y\_CR} \cdot A_i}{\sum A_i}$

Due to in-plane moment: $P_{x\_i}^{M_{z\_CR}} = -\frac{M_{z\_CR} \cdot A_i \cdot (Y_i - Y_{CR})}{I_p}$

$P_{y\_i}^{M_{z\_CR}} = \frac{M_{z\_CR} \cdot A_i \cdot (X_i - X_{CR})}{I_p}$



So,

5/5

$$\boxed{P_{sh\_i} = \sqrt{\left(P_{x\_i}^{F_{x\_CR}} + P_{x\_i}^{M_{z\_CR}}\right)^2 + \left(P_{y\_i}^{F_{y\_CR}} + P_{y\_i}^{M_{z\_CR}}\right)^2}}$$

Bolt group approach (cont) OFFPLANE CALCULATION → FZ, MX & MY

4/5 —— Axial forces in a given bolt i:

$$P_{z\_i} = P_{z\_i}^{F_{Z\_CR}} + P_{z\_i}^{M_{x\_CR}} + P_{z\_i}^{M_{y\_CR}}$$

Due to axial force: $P_{z\_i}^{F_{z\_CR}} = \frac{F_{Z\_\mathrm{CR}} \cdot K_{ti}}{\sum K_{ti}}$

Due to moment: $P_{z\_i}^{M_{x\_CR}} = \frac{M_{\mathrm{x\_CR}} \cdot \boldsymbol{K_{ti}} \cdot (Y_i - Y_{CR})}{I_x}$ $P_{z\_i}^{M_{y\_CR}} = -\frac{M_{\mathrm{y\_CR}} \cdot \boldsymbol{K_{ti}} \cdot (X_i - X_{CR})}{I_y}$

Where **Kti**, is the tension stiffness of the attachment. If no preload is considered then use Ai

~~5/5~~ → No component combination in offplane boltgroup as there is only tension load

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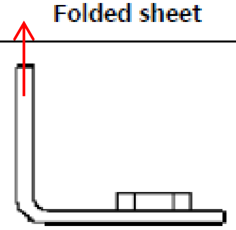
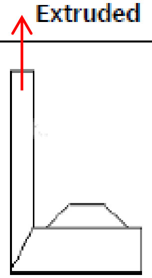
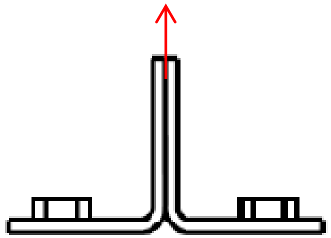
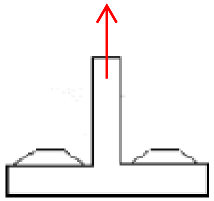
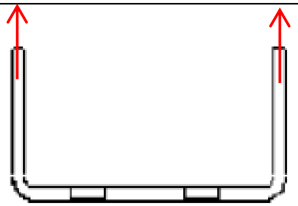
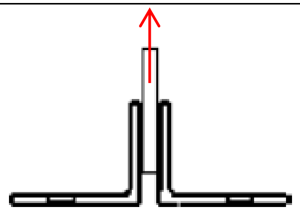
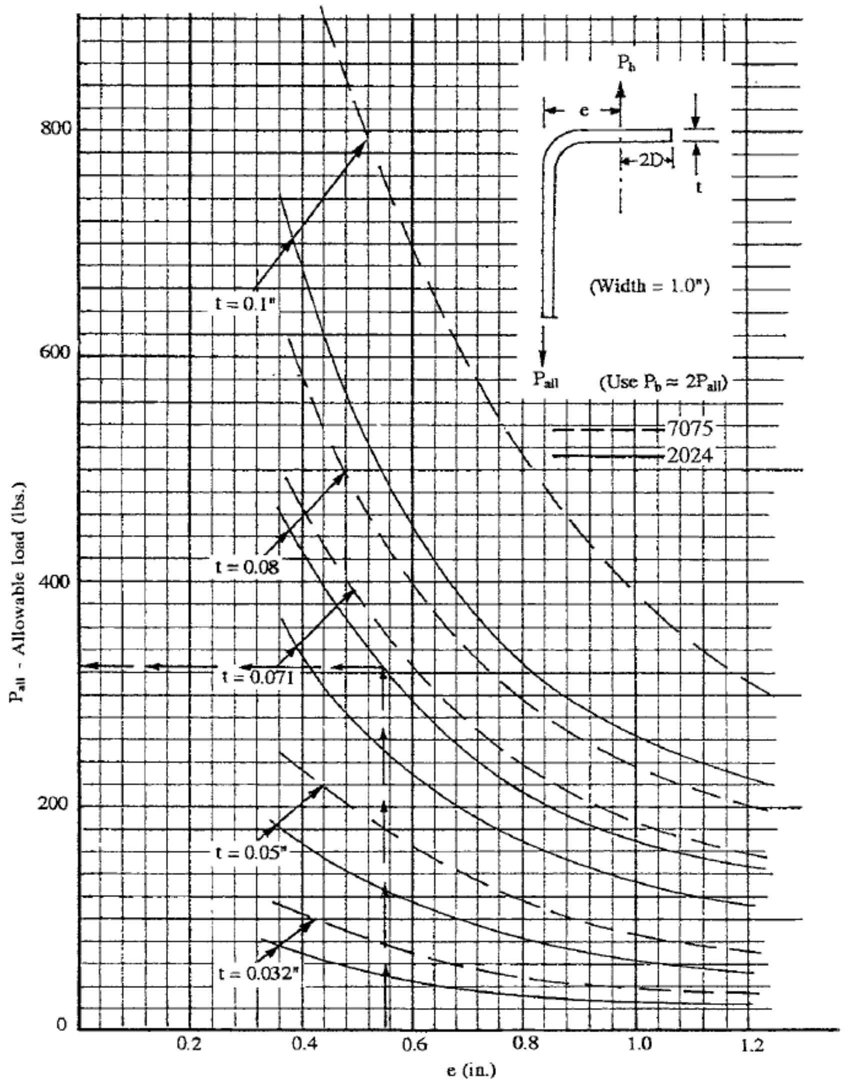
Tension Angles:

Designed to **support small loads** and manufactured from folded sheet profiles or extruded parts.

	Folded sheet	Extruded
L shaped profile (Simple Angle)		
T shaped profile (Tee Angle)		
Others	 U shaped profile	 Pair of angles

Tension Angles (cont):

Folded angles:

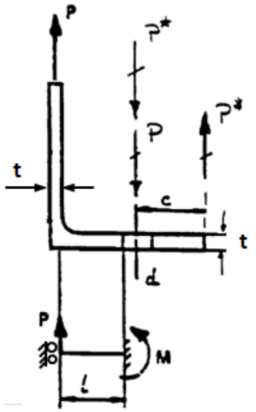


Tension **Angles** (cont):

Extruded/machined angles:

- Bending:

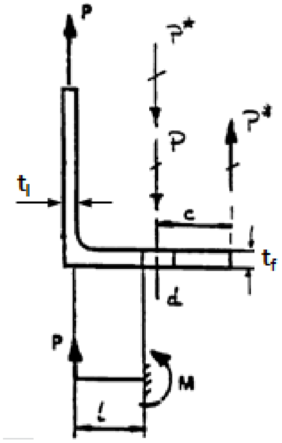
Web is able to support bending ($t_{web}>t_{base}$)



$$\boxed{M = \frac{P \cdot L}{2}}$$

Web is NOT able to support bending ($t_{web}<t_{base}$)

$$\boxed{M = P \cdot L}$$



$$\boxed{\sigma = \frac{6 \cdot \mathrm{M}}{\mathrm{b} \cdot t^2}}$$

Where,

M (Nmm): Maximum bending moment

P (N): Applied axial load (fitting load)

L (mm): Distance between bolt shank and the applied load line

b (mm): Bolt distance or effective width taking into account for flange bending analysis(≤4d)

t (mm): Foot thickness

Tension Angles (cont):

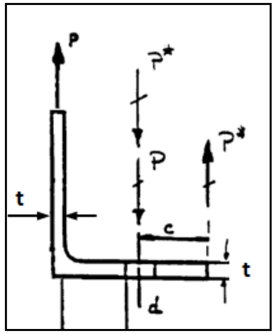
Extruded/machined angles:

— Bolt load:

$$P_b = \mathrm{P} + P^*$$

Where,

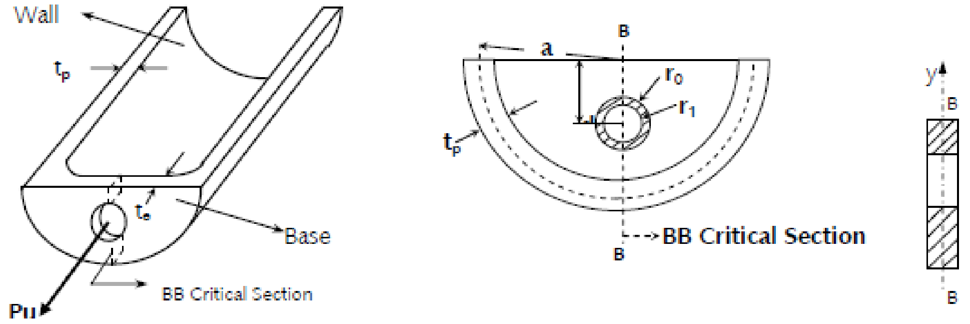
P (N): Applied load (fitting load)
P* (N): Load due to the prying effect



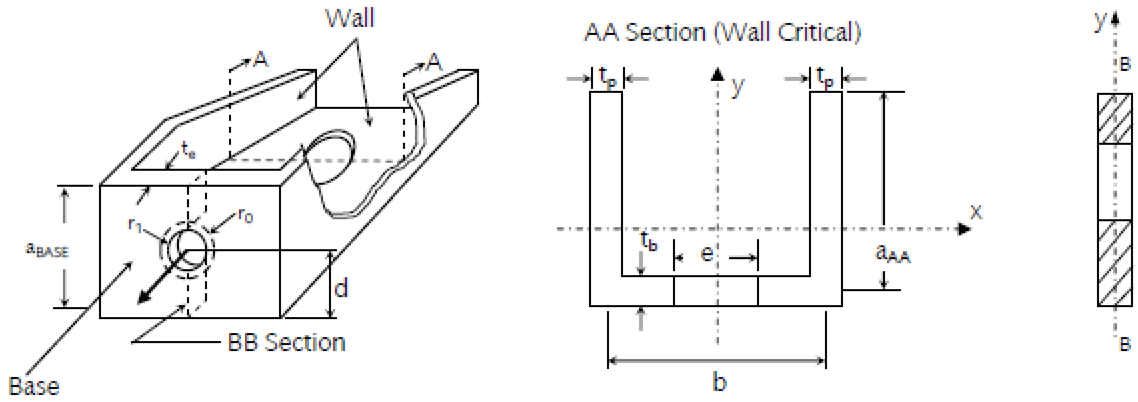
Fittings:

Designed to support **high loads**. Tension, Bending and Shear reserve factors are calculated for given loads on the base (base used to be the worst area of the fitting). Hereafter are presented 4 examples of fittings

1 Bath-up Fitting



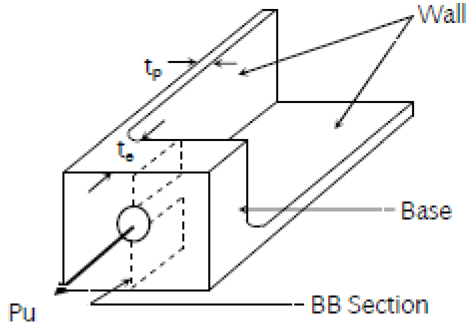
2 Channel Fitting

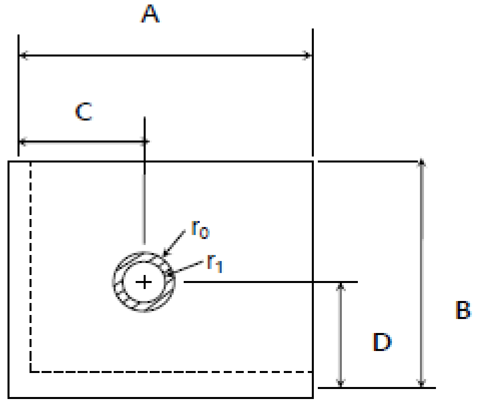


Fitting Analysis

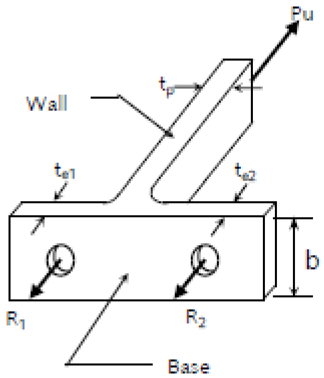
Fittings:

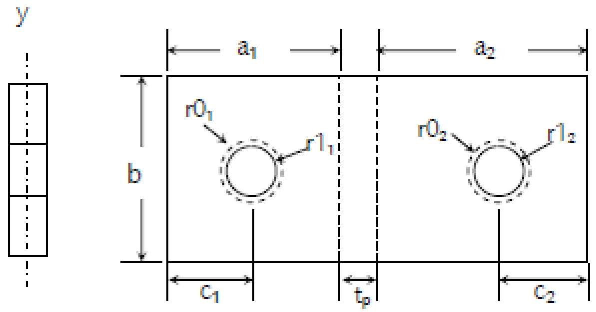
3 Angle Fitting





4 T-shaped Fitting

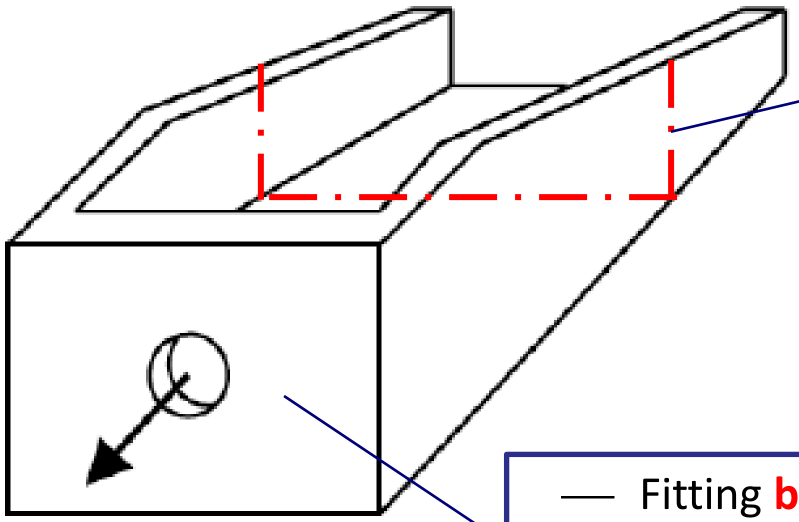




Fitting Analysis

Fittings (cont):

Failure modes:



- Fitting **wall** failure:
  - Tension strength
  - Shear strength
  - Tension-shear combination

- Fitting **base** failure:
  - Bending strength
  - Shear strength

Additionally, attachments should be analyzed

Fittings (cont):

Fitting **wall** cross-section failure:

— Tension: $$f_{tw} = \frac{P}{A}$$

Where,

P: Applied axial load

A: Net cross-sectional area (less fastener holes)

— Bending: $$f_{bw} = \frac{m_x \cdot y}{I_x} - \frac{m_y \cdot x}{I_y}$$

Where,

m: Moment of inertia about the neutral axis $m = P \cdot (d - c)$

P: Applied axial load

d: Distance between tension fastener and wall

c: Distance between neutral axis of the cross-section and wall

x,y: Distance between the extreme fibre and the neutral axis

I_x,I_y: Moment of inertia about the neutral axis

Fittings (cont):

Fitting **wall** cross-section failure (cont):

— Tension-bending combination: $\mathrm{f}_{wall} = f_{tw} + f_{bw}$

Limit: $RF_{wall,LL} = \frac{F_{ty}}{f_{wall,LL}}$

Ultimate: $RF_{wall,UL} = \frac{F_{tu}}{FF \cdot f_{wall,UL}}$

Fitting Analysis

Fittings (cont):

Fitting **base** cross-section failure:

— Bending: $$f_{be} = \frac{k \cdot P}{t_e^2}$$

Where,

k: Constant, see graphic

t_e: End thickness

Limit: $$RF_{\text{be},LL} = \frac{F_{ty}}{f_{\text{be},LL}}$$

Ultimate: $$RF_{\text{be},UL} = \frac{F_{bu}}{FF \cdot f_{\text{be},UL}}$$

Where,

F_{bu}: Allowable plastic bending stress with k=1,5 (rectangular shape)